

DATABASE DESIGN DOCUMENT

NAILVERSE

Identifying the Entities, Attributes Primary Keys and Relationships

Week - 2

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1. Introduction

This document defines the core entities, their attributes, the relationships between them, and the primary and foreign keys that form the database design for Nailverse. It serves as the blueprint for building the relational schema that supports product sales, seller and technician onboarding, service bookings, payments, shipments, and reviews.

2. Entities and Attributes

User (Customer)

Attribute	Description	Key
user_id	Unique identifier for each customer	PK
name	Full name of the customer	
email	Customer's email address	
phone_number	Customer's contact number	
password	Encrypted login password	
address	Delivery / billing address	
created_at	Account creation timestamp	

Seller

Attribute	Description	Key
seller_id	Unique identifier for each seller	PK
business_name	Registered store / business name	
owner_name	Name of the store owner	
email	Seller's email address	
phone_number	Seller's contact number	
store_address	Seller's business address	
verification_status	Verified / Pending status	
created_at	Registration timestamp	

Nail_Technician

Attribute	Description	Key
technician_id	Unique identifier for each technician	PK
name	Technician's full name	
email	Technician's email address	
phone_number	Technician's contact number	
certification_details	Certifications held by the technician	
portfolio_link	Link to past work / portfolio	
verification_status	Verified / Pending status	
seller_id	Links technician to the seller they work under	FK

Product

Attribute	Description	Key
product_id	Unique identifier for each product	PK
product_name	Name of the product	
category	Product category	
description	Short product description	
price	Selling price of the product	
stock_quantity	Units available in stock	
image_url	Path/link to the product image	
seller_id	Links product to the seller who lists it	FK

Service

Attribute	Description	Key
service_id	Unique identifier for each service	PK
service_name	Name of the nail service	
description	Short service description	
price	Cost of the service	
duration	Estimated duration in minutes	
technician_id	Links service to the technician who offers it	FK

Booking

Attribute	Description	Key
booking_id	Unique identifier for each booking	PK
booking_date	Date of the appointment	
time_slot	Scheduled time slot	
status	Confirmed / Rescheduled / Cancelled / Completed	
user_id	Links booking to the customer	FK
service_id	Links booking to the service booked	FK
technician_id	Links booking to the assigned technician	FK

Order

Attribute	Description	Key
order_id	Unique identifier for each order	PK
order_date	Date the order was placed	
total_amount	Total order value	
order_status	Placed / Shipped / Delivered / Cancelled / Returned	
user_id	Links order to the customer who placed it	FK

Order_Item

Attribute	Description	Key
order_item_id	Unique identifier for each order line item	PK
quantity	Number of units ordered	
price_per_unit	Price per unit at time of order	
order_id	Links item to its parent order	FK
product_id	Links item to the product ordered	FK

Cart

Attribute	Description	Key
cart_id	Unique identifier for each cart entry	PK
quantity	Number of units added	
added_date	Date the item was added to cart	
user_id	Links cart entry to the customer	FK
product_id	Links cart entry to the product	FK

Payment

Attribute	Description	Key
payment_id	Unique identifier for each payment	PK
amount	Amount paid	
payment_method	Card / UPI / Net Banking / Wallet	
payment_status	Pending / Success / Failed / Refunded	
transaction_date	Date and time of the transaction	
order_id	Links payment to an order (if applicable)	FK
booking_id	Links payment to a booking (if applicable)	FK

Shipment

Attribute	Description	Key
shipment_id	Unique identifier for each shipment	PK
courier_name	Logistics partner handling delivery	
tracking_number	Shipment tracking number	
shipment_status	Processing / Dispatched / In Transit / Delivered / Returned	
dispatch_date	Date the shipment was dispatched	
delivery_date	Date the shipment was delivered	
order_id	Links shipment to its order	FK

Review

Attribute	Description	Key
review_id	Unique identifier for each review	PK
rating	Rating value (1 to 5)	
comment	Written review text	
review_date	Date the review was submitted	
user_id	Links review to the customer who wrote it	FK
product_id	Links review to a product (if applicable)	FK
technician_id	Links review to a technician (if applicable)	FK

Admin

Attribute	Description	Key
admin_id	Unique identifier for each admin user	PK
name	Admin's full name	
email	Admin's email address	
password	Encrypted login password	
role	Admin role / access level	

3. Relationships

The table below summarizes how each entity connects to the others, along with the cardinality of each relationship.

Relationship	Entities Involved	Cardinality
Seller lists Product	Seller → Product	1 : M
Seller employs Technician	Seller → Nail_Technician	1 : M
Technician offers Service	Nail_Technician → Service	1 : M
User makes Booking	User → Booking	1 : M
Booking is for Service	Service → Booking	1 : M
Technician handles Booking	Nail_Technician → Booking	1 : M
User places Order	User → Order	1 : M
Order contains Products	Order ↔ Product (via Order_Item)	M : M
User adds to Cart	User → Cart	1 : M
Cart holds Product	Product → Cart	1 : M
Order has Payment	Order → Payment	1 : 1
Booking has Payment	Booking → Payment	1 : 1
Order has Shipment	Order → Shipment	1 : 1
User writes Review	User → Review	1 : M
Review is on Product	Product → Review	1 : M
Review is on Technician	Nail_Technician → Review	1 : M
Admin manages records	Admin → Seller, Product, Order	1 : M

4. Primary Keys & Foreign Keys Summary

Each entity's primary key uniquely identifies its records, while foreign keys establish the links between related entities.

Entity	Primary Key	Foreign Key(s)
User	user_id	—
Seller	seller_id	—
Nail_Technician	technician_id	seller_id → Seller
Product	product_id	seller_id → Seller
Service	service_id	technician_id → Nail_Technician
Booking	booking_id	user_id → User; service_id → Service; technician_id → Nail_Technician
Order	order_id	user_id → User
Order_Item	order_item_id	order_id → Order; product_id → Product
Cart	cart_id	user_id → User; product_id → Product
Payment	payment_id	order_id → Order; booking_id → Booking
Shipment	shipment_id	order_id → Order
Review	review_id	user_id → User; product_id → Product; technician_id → Nail_Technician
Admin	admin_id	—